2021 reconstruction of Italian user archetypes and energy groups

Dwelling probabilities coefficient matrix heating and electricity groups and assignment tables

4 November 2026

We applied the article's analysis sequence to the 2021 survey: household archetypes, dwelling-conditioned probabilities, polynomial regression, energy-group reduction and an Appendix A assignment matrix. Every 2021 statistical figure in this report is generated from that new run. The published 2013 results remain the historical reference.

Table 1 The published 2013 results and the 2021 reconstruction.

Output	2013 published result	2021 fitted result
Household archetype shares A1-A4	40.1 / 26.9 / 22.9 / 10.1%	24.20 / 32.87 / 21.76 / 21.16%
Heating groups	H1-H7	H1-H9
Electricity groups listed in Table 6	E1-E5	E1-E13
Heating-group mean range	4.67-9.67 h/day	6.17-11.11 h/day
Appliance electricity group mean range	1801-2895 kWh/year	1621-2843 kWh/year

In 2013, A1 was the largest published household type. In the 2021 fit, A2 is largest at 32.87%, while A4 accounts for 21.16%. The new grouping procedure produces nine heating groups and thirteen electricity groups. Group names are ordered by each year's fitted means; the same suffix denotes a rank within that year.

Table 2 Households contributing to the 2021 reconstruction.

Stage	Households
Raw 2021 release	54,069
Demographic fit	52,617
Dwelling probability stage	46,712
Heating-group stage	48,690
Electricity-group stage	52,617

The appendix covers all four locations, five 2021 dwelling categories and four user archetypes: 80 combinations and 160 H/E assignments. For example, a North West single-family dwelling occupied by user type A2 maps to H9 and E13 in the new table. Their group means are 11.11 h/day and 2843 kWh/year, respectively.

Households contribute equally to the primary clustering and empirical cell ratios; the polynomial gives equal weight to each of the 21 fitting windows. Separate PESO-weighted shares, group means and probability regressions are supplied. The 2021 apartment categories describe floor position; the 2013 apartment categories describe building size. Appliance electricity is an ownership/use model with fixed reference coefficients.

Definition of user archetypes

The 2021 four-cluster fit uses the article's loneliness and generation coordinates and an age-adjusted schooling index. Its labels are aligned with the historical interpretations. In A1, mean maximum age is 81.2 years, compared with the paper's 78.3. In A2, mean household size is 3.2 members, compared with 3.7 in 2013. Table 3 gives the corresponding profile summaries.

Table 3 User archetypes derived from k-means clustering.

2013 published values. Average (standard deviation).

cluster	age min	age max	n members	education
1	70.2 (11.2)	78.3 (5.0)	1.8 (0.7)	0.5 (0.1)
2	21.2 (7.6)	57.2 (8.6)	3.7 (0.9)	0.7 (0.1)
3	46.0 (12.8)	63.0 (8.0)	2.3 (0.8)	0.5 (0.1)
4	55.6 (17.1)	72.2 (9.0)	2.1 (0.8)	0.8 (0.1)

2021 fitted values. Average (standard deviation).

cluster	age min	age max	n members	education
1	71.6 (15.3)	81.2 (10.7)	1.7 (0.7)	0.3 (0.1)
2	17.3 (12.3)	49.1 (10.3)	3.2 (1.1)	0.8 (0.1)
3	31.0 (17.0)	50.8 (10.7)	2.3 (1.1)	0.5 (0.1)
4	56.2 (17.1)	71.5 (9.4)	2.0 (0.9)	0.6 (0.1)

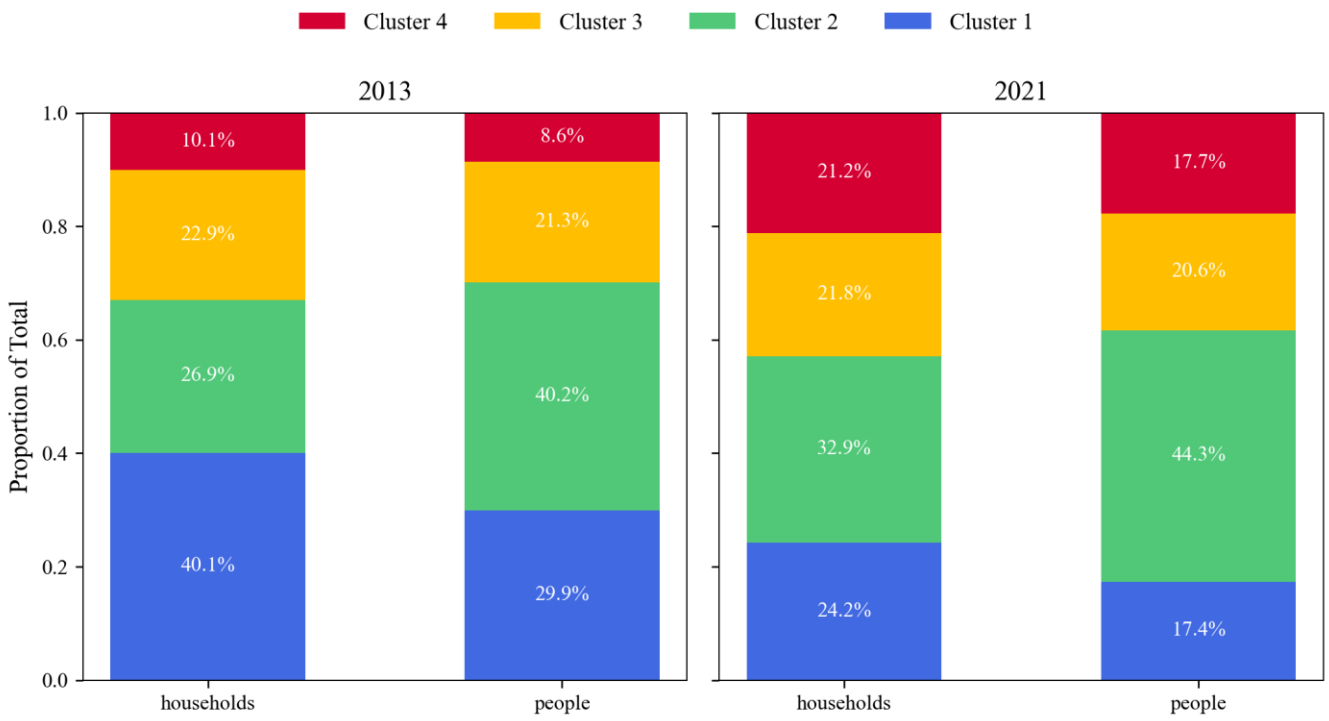


Fig. 5. Proportion of each cluster among households and people.

The 2013 panel retains the printed shares. The 2021 panel uses the new 52,617-household fit. Its survey-weighted household shares are 28.31%, 27.70%, 27.65% and 16.35%. The schooling and age assumptions are stated on page 10; all 2021 outputs use these same primary labels.

Associating user archetypes to building types

We reconstructed windowed empirical probabilities using the article's 20-year building-age grid, 40 m² area grid and centred half-windows of 6 years and 8 m². The eligible dwelling cohort has 46,712 records; 12,334 fall inside at least one of the 63 grid windows. Fifty-seven windows have at least five observations. The primary polynomial uses the 21 supported windows centred at ages 20-140 years and areas 40, 80 and 120 m².

Let a be building age in years and s dwelling area in m². The fitted 2021 matrix, rounded to five decimal places, is

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} 0.06496 & 0.54972 & 0.00000 & 0.00000 & -0.26970 & 0.00000 \\ 0.39519 & -0.55058 & 0.10305 & 0.00000 & 0.30849 & 0.00000 \\ 0.45712 & -0.08898 & -0.21347 & 0.00000 & 0.00000 & 0.00000 \\ 0.15278 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.07919 \end{bmatrix} \begin{bmatrix} 1 \\ a / 100 \\ s / 100 \\ a\,s / 10000 \\ a^2 / 10000 \\ s^2 / 10000 \end{bmatrix} + \varepsilon$$

The matrix product is the fitted part; ε denotes the four residuals. The full-precision coefficient CSV is authoritative. Basis order is 1, a/100, s/100, a s/10000, a²/10000, s²/10000. The six-term starting basis follows the article's matrix; backward elimination retains terms at p≤0.05, with the intercept retained.

Table 4 Regression results.

Variable	Prob 1 coefficient	p-value	Prob 2 coefficient	p-value	Prob 3 coefficient	p-value	Prob 4 coefficient	p-value
intercept	0.06496	0.184	0.39519	<0.001	0.45712	<0.001	0.15278	<0.001
age/100	0.54972	<0.001	-0.55058	<0.001	-0.08898	0.006	0	-
area/100	0	-	0.10305	0.008	-0.21347	<0.001	0	-
age × area/10000	0	-	0	-	0	-	0	-
age²/10000	-0.26970	0.004	0.30849	0.001	0	-	0	-
area²/10000	0	-	0	-	0	-	0.07919	0.002
F-statistic	13.98		9.34		23.72		13.54	
Adjusted R^2	0.565		0.556		0.694		0.385	
RMSE	0.0484		0.0459		0.0482		0.0495	

The selected 2021 equation relates A1 mainly to building age, A2 to age and area, A3 to a linear combination of age and area, and A4 to squared area. This changes the fitted shapes from the 2013 equation while preserving the same six-column representation.

At the 21 fitting centres, individual estimates lie within 0-1 and their four-value sums range from 0.98044 to 1.03517. On the full 63-point grid, 35 points have an individual estimate outside 0-1, all beyond the primary area domain. The saved grid preserves those raw range and sum flags. The following probability figures use the primary fit domain.

Dwelling probability profiles

In the 2021 equation, A2 probability rises with dwelling area and A3 probability falls with area at a fixed building age. A1 changes with building age. After variable selection, the A4 curve depends on area. Figure 6 places the empirical proportions and fitted curves on the article's four-panel axes.

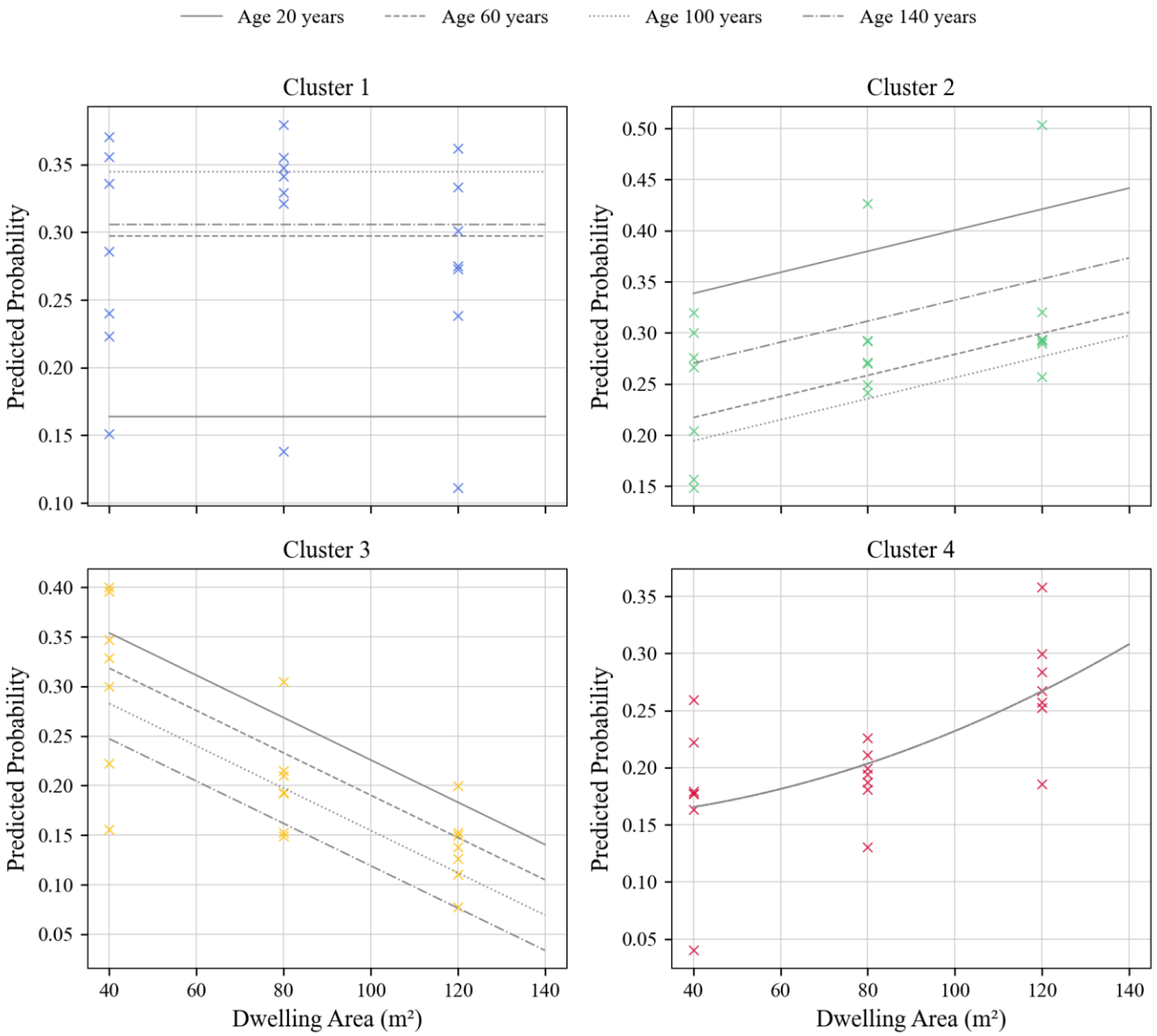


Fig. 6. Probability of each cluster of users as the dwelling's age changes. Points are calculated proportions. Lines are predictions of the regression model.

Crosses are the 21 empirical fitting windows. Gray lines correspond to building ages 20, 60, 100 and 140 years over areas 40-140 m². In Cluster 4 the selected equation gives coincident age curves. The horizontal axis is dwelling area, matching the original figure's layout.

The full empirical grid and its household counts are exported for inspection. The survey-weighted grid and a separately fitted weighted coefficient matrix are also available, so the weighting choice can be examined without changing the paper-style primary fit.

Alignment between fitted and empirical probabilities

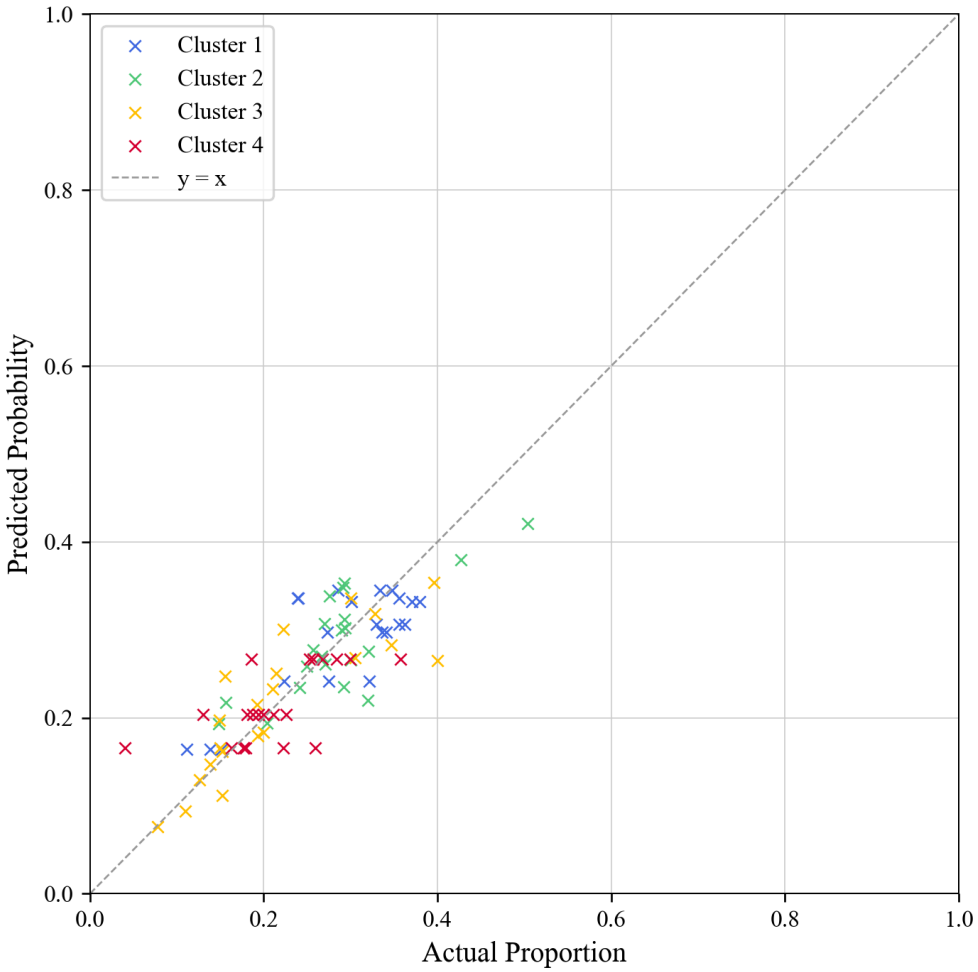


Fig. 7. Alignment Between Predicted and Actual Values Across Clusters.

The plot uses the 21 fitting windows. Crosses retain the paper's four cluster colours; the dashed line is y=x. These are apparent fitted values. The household holdout scores below provide a separate diagnostic conditional on the demographic fit.

Table 4a Published and new probability regression fit summaries.

Type	2013 Adjusted R^2	2021 Adjusted R^2	2013 RMSE	2021 RMSE	2021 holdout RMSE
A1	0.94	0.565	0.020	0.0484	0.0719
A2	0.93	0.556	0.016	0.0459	0.0893
A3	0.87	0.694	0.008	0.0482	0.0983
A4	0.84	0.385	0.011	0.0495	0.0511

The 2013 paper reports adjusted R^2 values of 0.94, 0.93, 0.87 and 0.84. In the new 2021 window regressions, they are 0.565, 0.556, 0.694 and 0.385. The new residuals are larger, and the household-held-out RMSEs range from 0.0511 to 0.0983. Thus the 2021 reconstruction shows a less tight relationship between the two dwelling predictors and the fitted user types on its supported windows.

Deriving energy patterns for specific occupant groups

The energy stage starts from location × dwelling type × user archetype cells. Each response has 80 observed starting cells. We successively merge the pair with the largest Tukey-Kramer adjusted p-value, then order the resulting groups by their response mean. The 2021 reduction makes 71 heating merges and 67 electricity merges, producing H1-H9 and E1-E13.

Table 5 ANOVA results (single variables and interactions).

Term	Heating F-statistic	p-value	Electricity F-statistic	p-value
Usercluster	43.27	<0.001	2090.12	<0.001
Dwellingtype	103.18	<0.001	82.95	<0.001
Location	1225.45	<0.001	83.57	<0.001
Age dwel	45.02	<0.001	88.92	<0.001
Area dwel	194.41	<0.001	4853.78	<0.001
CoolingSystemUse	0.68	0.410	680.82	<0.001
Usercluster × Dwellingtype	5.01	<0.001	5.62	<0.001
Usercluster × Location	8.72	<0.001	4.03	<0.001
Dwellingtype × Location	11.01	<0.001	7.11	<0.001
Age dwel × Area dwel	13.86	<0.001	1.31	0.253

Type-II OLS ANOVAs use 43,569 heating records and 46,712 electricity records with known construction age and area. The original subjective insulation question is absent in 2021, so insulation terms are omitted. The corresponding heating model has R^2=0.111; the electricity model has R^2=0.266. Regional and dwelling factors retain their 2021 definitions.

Location has the largest heating F-statistic among the main categorical factors. The user-archetype effect is pronounced in the appliance electricity model. Interactions among user type, dwelling type and location also contribute to the fitted response patterns. The group procedure below uses those three cell identifiers for both responses, following the final saved group structure and the published Appendix A layout.

The final partition contains nine heating groups with means from 6.17 to 11.11 h/day, compared with seven published groups from 4.67 to 9.67 h/day in 2013. It contains thirteen modelled appliance-electricity groups with means from 1621 to 2843 kWh/year, compared with the five groups listed in the paper from 1801 to 2895 kWh/year.

The iterative procedure reuses the same observations as groups are merged. Its terminal pairwise tests describe the selected partition. The implementation uses full-precision p-values with a deterministic tie rule. Group labels are ranks within 2021 and are linked to their actual cell memberships in Appendix A.

Heating and electricity group summaries

The following means and standard deviations come from the new 2021 household assignments. Each example is an observed location, dwelling and user-type cell belonging to that group. The complete assignments appear in Appendix A.

Table 6a Significant groups for heating system usage.

Group	Example	Average (SD) h/day	n
H1	A2, intermediate-floor flat, South/Islands	6.17 (3.83)	7,581
H2	A2, intermediate-floor flat, Central	6.84 (4.02)	6,128
H3	A2, multi-family house, South/Islands	7.75 (4.47)	7,681
H4	A3, single-family house, South/Islands	8.36 (4.66)	2,159
H5	A2, intermediate-floor flat, NW	8.88 (5.02)	7,773
H6	A3, multi-family house, NW	9.61 (5.26)	2,566
H7	A1, intermediate-floor flat, NW	10.19 (5.24)	6,032
H8	A2, multi-family house, NE	10.56 (5.39)	4,706
H9	A1, single-family house, NE	11.11 (5.68)	4,064

Table 6b Significant groups for annual electricity consumption

Group	Example	Average (SD) kWh/year	n
E1	A1, intermediate-floor flat, South/Islands	1621 (518)	5,184
E2	A3, intermediate-floor flat, NW	1728 (613)	3,443
E3	A1, single-family house, South/Islands	1791 (641)	5,645
E4	A3, intermediate-floor flat, South/Islands	1854 (649)	3,893
E5	A4, intermediate-floor flat, South/Islands	1906 (651)	7,654
E6	A1, single-family house, NE	2027 (717)	2,596
E7	A4, single-family house, South/Islands	2136 (752)	3,304
E8	A2, intermediate-floor flat, Central	2228 (781)	6,610
E9	A2, intermediate-floor flat, South/Islands	2295 (794)	5,452
E10	A2, top-floor flat, Central	2362 (831)	2,310
E11	A2, single-family house, South/Islands	2527 (854)	1,981
E12	A2, multi-family house, NW	2684 (842)	2,322
E13	A2, multi-family house, NE	2843 (893)	2,223

NE=North East; NW=North West. Sample sizes are unweighted. PESO-weighted group means are supplied in separate CSVs. Electricity includes the audited appliance-model components and reference assumptions described on page 11.

Daily heating on time and annual appliance electricity

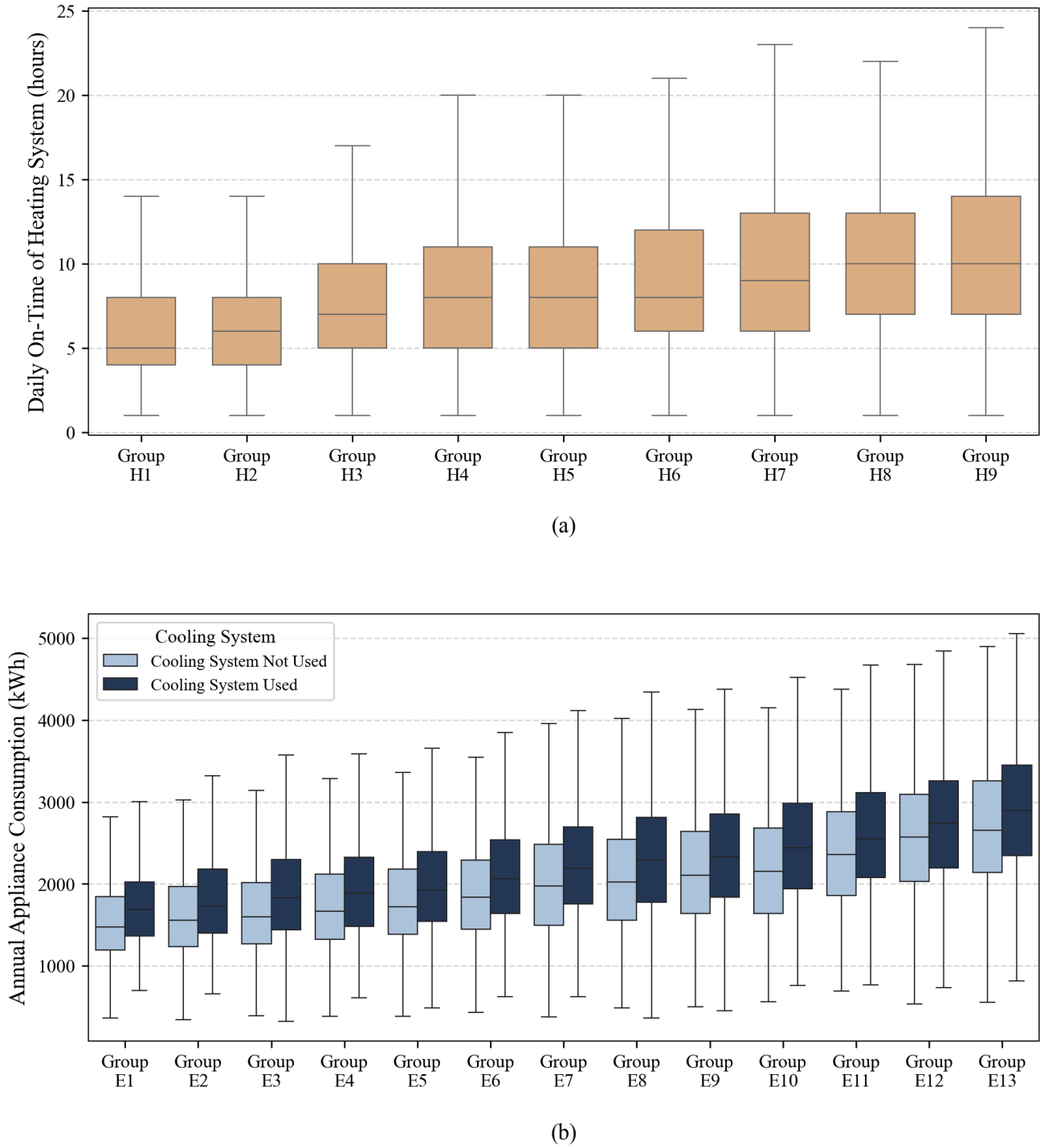


Fig. 8. Variation of (a) daily heating system operating hours and (b) annual electricity consumption based on significant groups of dwelling types, geographical location, and user type.

All boxes use the new 2021 assignments. Declared heating hours refer to cold months of 2020. Heating shows H1-H9 without an insulation split because that subjective rating is absent in the 2021 release. Electricity shows E1-E13 split by declared cooling use. Boxes span the 25th-75th percentiles, with medians and non-outlying whiskers; outliers are hidden. The displayed outcomes and group axes follow the paper.

Appendix A

Figure A.1. Assignment of user groups based on building type, location and user archetype.

Location	Dwelling Type	Heating Group				Electricity Group			
		User Cluster 1	User Cluster 2	User Cluster 3	User Cluster 4	User Cluster 1	User Cluster 2	User Cluster 3	User Cluster 4
North East	Apartment on ground, semi-basement or raised floor	H9	H5	H5	H6	E1	E10	E5	E5
	Apartment on intermediate floor	H8	H5	H5	H7	E2	E8	E3	E4
	Apartment on top floor	H7	H6	H5	H8	E2	E9	E4	E5
	Single Family House	H9	H9	H7	H9	E6	E13	E8	E9
	Multi Family House	H8	H8	H7	H8	E5	E13	E8	E8
North West	Apartment on ground, semi-basement or raised floor	H7	H6	H5	H7	E1	E9	E3	E3
	Apartment on intermediate floor	H7	H5	H5	H8	E1	E8	E2	E5
	Apartment on top floor	H7	H6	H5	H8	E1	E10	E3	E5
	Single Family House	H9	H9	H7	H9	E5	E13	E7	E10
	Multi Family House	H7	H7	H6	H8	E4	E12	E7	E8
Central	Apartment on ground, semi-basement or raised floor	H2	H3	H1	H3	E1	E8	E3	E5
	Apartment on intermediate floor	H2	H2	H1	H3	E2	E8	E3	E4
	Apartment on top floor	H3	H2	H2	H3	E2	E10	E5	E5
	Single Family House	H4	H5	H3	H4	E5	E12	E7	E9
	Multi Family House	H3	H3	H3	H3	E5	E12	E7	E7
South and Islands	Apartment on ground, semi-basement or raised floor	H2	H2	H1	H1	E1	E8	E2	E3
	Apartment on intermediate floor	H1	H1	H1	H1	E1	E9	E4	E5
	Apartment on top floor	H2	H2	H1	H2	E1	E9	E3	E5
	Single Family House	H5	H5	H4	H3	E3	E11	E6	E7
	Multi Family House	H4	H3	H3	H3	E3	E11	E6	E6

New 2021 fitted lookup. The three apartment rows follow the released ground/semi-basement/raised, intermediate and top-floor categories. Group means are in Table 6. H and E labels are within-year ranks. All 160 assignments are also supplied as long and wide CSV tables, and as standalone SVG and PNG graphics.

Model construction and sensitivity

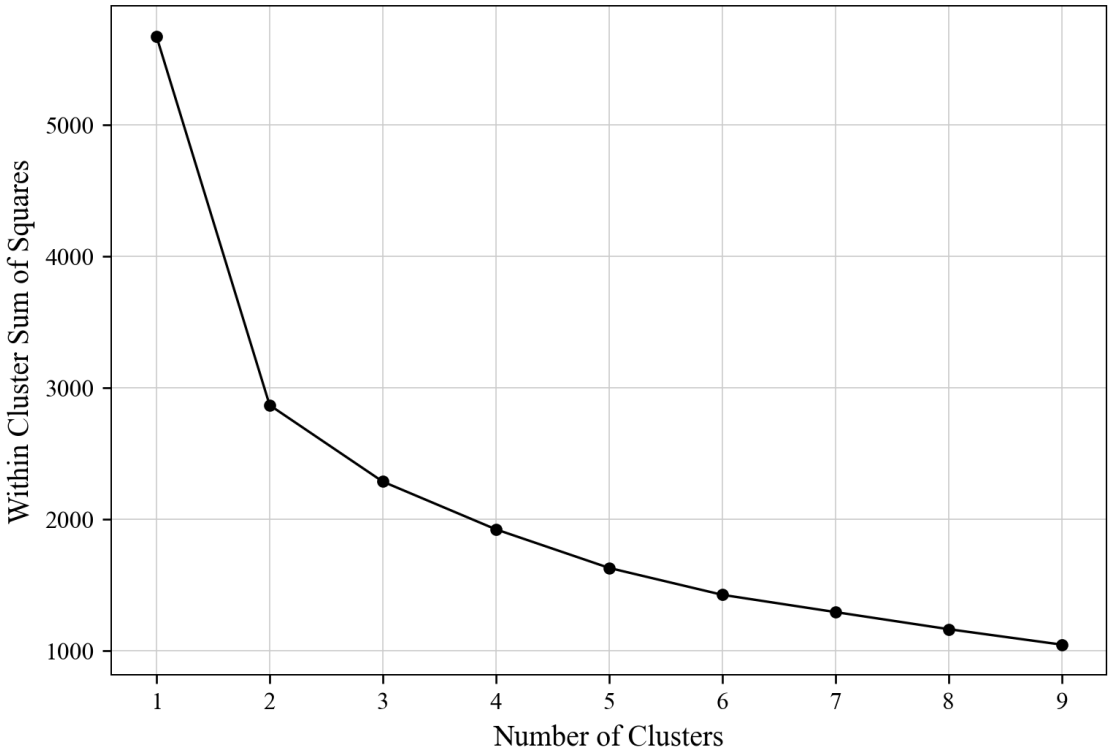


Fig. 3. Within Cluster Sum of Squares as a function of number of clusters.

The fresh 2021 fit uses K=4 to retain the article's four-type resolution. K-means uses 20 initializations, seed 42, a 500-iteration limit and tolerance 10^-8. Seed 42 draws integer ages uniformly within released intervals, bounding ages 80+ at 100. Loneliness and generation use a new 2021 min-max transform; the bounded schooling index is used directly. An orthogonal full PCA is recorded. Semantic labels are aligned to historical reference coordinates. All original 2013 assignments remain fixed.

The released 0-10 age groups have blank schooling. Their ages and member counts are retained, while these blank entries are omitted from the schooling mean. For members with supported schooling, the index uses education years divided by age-specific potential schooling, capped at one. Primary coding assigns five years to the merged no-credential/primary category and twelve years to the upper-secondary category.

Table 7 Archetype shares under schooling assumptions.

Coding	A1	A2	A3	A4
Primary	24.20%	32.87%	21.76%	21.16%
Lowest schooling category at 0 years	15.93%	28.01%	26.39%	29.67%
Upper-secondary category at 13 years	22.64%	35.19%	21.21%	20.97%

Primary demographic coverage is 52,617 households, including 7,737 with a released child aged 0-10. Questionnaire F.10 routes schooling to age 10 or older and relationship codes 1-5. The primary fit excludes 903 otherwise size-eligible profiles with unsupported schooling and 549 households with 6+ members. The lowest schooling-category assumption materially affects the fitted shares; all H/E and PFE outputs use the declared primary coding.

Sources and reproducibility

The analysis follows the saved article code and the final published output structure. Family features trace to 02_Families_Analysis.py; the four-type fit to 01_PCA_Exploratory.py and the clustering representation; dwelling windows and polynomial regressions to the 05-series files and pfe_pipeline.py; energy-group reduction and Figure 8 to 07_ANOVA_analysis.py and the recovered iterative grouping function. The final saved electricity input and published Appendix A determine the three-factor cell structure.

The legacy files contain exploratory cells with different K values, restricted regression trials and two-factor electricity trials. The new run records the selected article structure: four demographic types, the six-column polynomial basis and location × dwelling × user cells. The new Tukey implementation uses full-precision p-values and deterministic ties. It writes new group memberships and lookup tables while leaving every 2013 source and result unchanged.

Energy input assumptions

The annual appliance outcome is rebuilt from 2021 ownership and use using the audited reference model. It includes lighting, small appliances, refrigeration, washing, drying, dishwashing, screens, computers and standby. Reference power and efficiency parameters are held fixed. Missing bulb-hour, small-appliance and refrigerator-detail fields use documented 2013-derived reference assumptions, with lower and upper model scenarios saved in the input package. The 2013 component reconstruction agrees with the stored reference totals within 1.82 × 10^-12 kWh.

Unknown construction age is kept out of the probability and age/area ANOVA stages and is retained for H/E grouping. Heating uses frequent-user responses with all three valid daily blocks. Cooling use is established from declared frequencies or explicit absence. The regional expansion weights are PESO; weighted diagnostics are separate from the equal-household fits. Statistical tests describe these selected survey/model records, with survey-mode, reference-period and imputation differences recorded in the methods files.

Primary sources

Khajedehi, Vivian and Zarrella (2025). Identification of Italian user archetypes for building energy simulations based on a national survey. Energy and Buildings 344, 115924.

ISTAT. Indagine sui consumi energetici delle famiglie. Native 2021 microdata, record layout, classifications and questionnaire.

Analysis package

The output/article_method_2021_20261104 folder contains audited inputs, primary household labels, empirical and weighted probability grids, coefficient matrices, fit and holdout diagnostics, H/E merge histories and terminal tests, household assignments, group summaries and all Appendix A cells. Figure outputs include SVG, PNG and PDF versions. The report supplies the numerical equation as editable Word mathematics and the appendix as an editable Word table.

The independent review reproduces the raw-data cohort, probability calculations and regressions, all H/E merges, ANOVAs and Appendix assignments. The final manifest records protected-source integrity, package versions and output hashes. The original 2013 baseline and prior reports are retained.